Sharif University of Technology

What is Probability?

A Review of Philosophical Theories of Probability

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Overview of the Lectures

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- **The Frequency Theory**: Probability of an outcome is the limiting frequency with which that outcome appears in a long series of similar events.
- **The Propensity Theory**: Probability is a propensity inherent in a set of repeatable conditions.
  - To say that the probability of a particular outcome is p is to claim that the repeatable conditions have a *propensity* such that, if they were to be repeated a large number of times, they would would produce a *frequency* of the outcome close to p.

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- I will focus on Cambridge in the Edwardian era before the First World War, and, in particular, on the work of Keynes.
- Keynes wrote *Treatise on Probability* in 1921.

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- Keynes in his 1938 talk *My Early Beliefs*:

I was writing under the joint influence of Moores Principia Ethica and Russells Principia Mathematica.

The Logical Theory

- Moores argument in *Principia Ethica*:

We should act in order to bring about the greatest amount of goodness, but we can only calculate the probable effects of our actions in the immediate future. We really know nothing about their long-term consequences. Moreover, these long-term consequences may be such as to reverse the balance of good produced by our action in the short term.

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- Keynes thought that the mistake (as he saw it) in Moores argument lay in Moores adopting the wrong interpretation of probability.

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- Given this indifference we should assign possibilities (a) and (b) equal probabilities.
- The desire to maximise *expected* good now leads us to prefer action A.
- The general conclusion is that we should carry out the action which produces the most goodness in the short term, *even if this contradicts the rules of conventional morality*.

Probability as a logical relation

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- Now consider the following problem:
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- Hume argued, and this is in agreement with modern logic, that neither h nor d follow logically from e.
- Could we not say that e *partially* entails h and d, since e surely gives some support for these conclusions?
- This line of thought suggests that there might be *a logical theory of partial entailment* which **generalises** the ordinary theory of full entailment which is found in deductive logic.

Partial Entailment

- Keynes (1921):

  Inasmuch as it is always assumed that we can sometimes judge directly that a conclusion follows from a premiss, ***it is no great extension*** *of this assumption to suppose that we can sometimes recognise that a conclusion* ***partially follows from****, or* ***stands in a relation of probability*** *to a premiss.*

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- So a probability is the degree of a **partial entailment**.

Are all probabilities conditional?

- One immediate consequence of this approach is that it makes all probabilities conditional. Keynes (1921):

> *No proposition is in itself either probable or improbable, just as no place can be intrinsically distant;* ***and the probability of the same statement varies with the evidence presented****, which is, as it were, its origin of reference.*

Degree of Rational Belief

- Keynes (1921):

> *Let our premisses consist of any set of propositions e, and our conclusion consists of any set of propositions h, then, if a knowledge of e justifies a rational belief in a of degree α, we say that there is a probability-relation of degree α between e and h.*

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- Here Keynes makes the assumption that if e partially entails e to degree α, then given e it is *rational* to believe h to degree α.

Degree of Rational Belief

This idea has been challenged by Popper and Carnap:

- Suppose we have finite evidence and a generalisation which may have a potentially infinite number of instances. Now here h goes so to speak *infinitely beyond* e:
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  Popper argues, the degree to which e partially entails h is zero.
- Poppers argument continues, although the degree to which finite evidence partially entails a universal generalisation is zero, it may nonetheless **be possible to have a non-zero degree of rational belief** in a universal generalisation given finite evidence.

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- Popper accepts a logical interpretation of probability where probability is identified with degree of partial entailment.
- Popper (1959):

> *we may learn from experience more and more about universal laws* ***without*** *ever increasing their probability; ... we may test and corroborate some of them better and better, thereby increasing their degree of corroboration without altering their probability* ***whose value remains zero.***

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Keynes' Solution:

- On the general problem of knowledge Keynes adopted a *Russellian* position. Russell held that **some of our knowledge is obtained directly or *by acquaintance***.
- His views on what we could know in this way varied, but the set always included **our immediate sense perceptions**.
- The rest of our knowledge is **knowledge by description** and is ultimately based on our **knowledge by acquaintance**.

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- Indeed, Keynes appears to argue at times that all logical relations are known by direct acquaintance. Thus he says:

  When we know something by argument this must be through direct acquaintance with some logical relation between the conclusion and the premiss.

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- Keynes does indeed recognise this:

> *While we may possess a faculty of* ***direct recognition*** *of many relations of probability, as in the case of many other logical relations, yet* ***some may be much more easily recognisable than others****. The object of a logical system of probability is to enable us to know the relations, which cannot be easily perceived, by means of other relations which we can recognise more distinctly to convert, in fact, vague knowledge into more distinct knowledge.*

- Some of the axioms which Russell and Whitehead used in Principia Mathematica were far from being obviously correct to the intuitions of many mathematicians, and, as we shall in the future, similar criticisms were directed against Keynes's Treatise.

Rational Belief or Belief

- For Keynes probability was degree of rational belief not simply degree of belief.

> $\cdots$ *in the sense important to logic, probability is* ***not*** *subjective. It is not, that is to say, subject to human caprice.* ***A proposition is not probable because we think it so****. When once the facts are given which determine our knowledge, what is probable or improbable in these circumstances has been fixed objectively, and* ***is independent of our opinion*** $\cdots$

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- He writes:

  The perceptions of some relations of probability may be outside the powers of some or all of us.

Measurable probabilities

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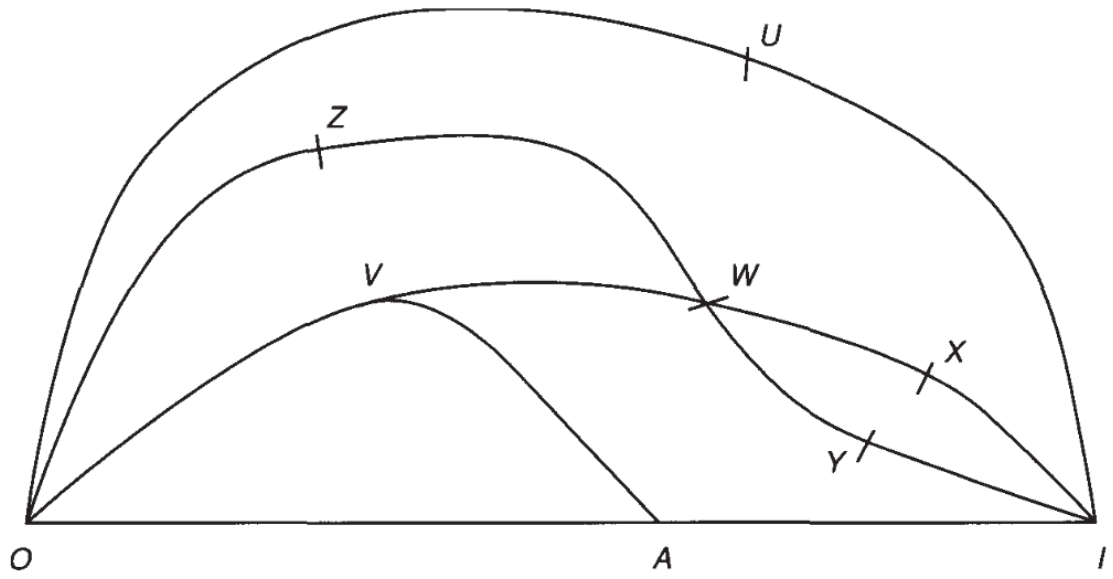

Issues with Keynes' Theory

- De Finetti regarded Keyness non-numerical probabilities as an *unfortunate departure* from the **simplicity** and **power** of the mathematical theory of probability. He wrote:

  Keyness position is certainly not suited to the development of a mathematical probability theory and is also hardly in keeping with the intuitive idea of probability. I myself regard as unacceptable, as a matter of principle, Keyness position (the more so since the reservations which he had disappear when one adopts a subjective point of view).

Principle of Insufficient Reason

- What then are the cases in which numerical values can be assigned to probabilities? Keynes answers:

  In order that numerical measurement may be possible, we must be given a number of equally probable alternatives.

- He even claims that this is something on which all probabilists agree:

  It has always been agreed that a numerical measure can actually be obtained in those cases only in which a reduction to a set of exclusive and exhaustive equiprobable alternatives is practicable.

Principle of Insufficient Reason

- Keynes gives the following preliminary statement of the principle:

  The Principle of Indifference asserts that if there is no known reason for predicating of our subject one rather than another of several alternatives, then relatively to such knowledge the assertions of each of these alternatives have an equal probability.

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- **Bayes Law**:

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- $h \in \{h_1, \ldots, h_m\}$ and uniform!
- $\mathbb{P}(e) = \sum_{i=1}^{m} \underbrace{\mathbb{P}(h_i)}_{\frac{1}{m}} \mathbb{P}(e|h_i)$.

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At the core of each paradox, lies the following problem:

$$X \sim \text{Unif}(0, 2) \text{ or } X^2 \sim \text{Unif}(0, 4)$$

What comes next?

In the next session, we will start discussing the *Subjective Theory* of probability as a way to resolve issues we encountered when studying Logical probabilities.